

Fix C++26 by optimizing `linalg::conjugated` for noncomplex value types

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Contents

- [1 Authors](#)
- [2 Revision history](#)
- [3 Abstract](#)
- [4 Presentation](#)
 - [4.1 Terms](#)
 - [4.2 Common practice: “conjugate transpose of a noncomplex matrix is just the transpose”](#)
 - [4.3 `conjugated`, `transposed`, and `conjugate\_transposed` views](#)
 - [4.4 How does `conjugated` work currently?](#)
 - [4.5 This is correct, but can hinder optimization](#)
 - [4.6 Fix: `conjugated\(A\)` should return `A` if `A` is noncomplex](#)
- [5 Design justification](#)
 - [5.1 Introduction](#)
 - [5.2 Current behavior of `conjugated`](#)
 - [5.3 Why change the current behavior?](#)
 - [5.4 P1673 layouts and accessors are not “just tags”](#)
 - [5.5 Change: `conjugated\(x\)` may no longer have `const element\_type`](#)
 - [5.6 What if the input `mdspan` has `conjugated\_accessor` with noncomplex `element\_type`?](#)
- [6 Implementation](#)

— 7 Acknowledgments

— 8 Wording

§ 1 Authors

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§ 2 Revision history

- Revision 0 was submitted for the post-Kona mailing on 2023/11/15.
- Revision 1 will be submitted for the post-Tokyo mailing on 2024/04/16.
 - Add explanation to Wording section (actual wording change is not affected)
 - Minor revisions of non-wording material
 - Add link to implementation
 - Change title and abstract, to emphasize that delaying this until after C++26 would be a breaking change
- Revision 2 will be submitted by 2024/08/15.
 - Minor wording fix (define “E” in “conj(E)”)
 - Bump value of `__cpp_lib_linalg` macro
 - Add nonwording “Presentation” section

§ 3 Abstract

We propose the following change to the C++ Working Paper. If an `mdspan` object `x` has noncomplex `value_type`, and if that `mdspan` does not already have accessor type `conjugated_accessor<A>` for some nested accessor type `A`, then we propose to change `conjugated(X)` just to return `X`. Delaying this until after C++26 would be a breaking change.

§ 4 Presentation

Reviewers who are not familiar with `std::linalg` might like to start with this section. It summarizes why `conjugated` exists, how it works, and why its definition needs to change.

§ 4.1 Terms

— A complex number $z = x + iy$ has a *real part* x and an *imaginary part* y .

- Mathematical convention calls x the “real part” even if x isn’t necessarily a real number (e.g., it could be an integer).
- In `std::linalg`, a “complex number” is any number type, not necessarily `std::complex`, where `conj` is ADL-findable. (Users define their own complex number types to work around various `std::complex` issues, as P1673 explains.)
- For numbers that are not complex, we say “noncomplex” and not “real” because `std::linalg` does not require them to be real numbers, or even necessarily to be of arithmetic types (e.g., they could be user-defined number types).
- The *conjugate* of a complex number $z = x + iy$ is $x - iy$.
- The conjugate of a noncomplex number is just the number (as its imaginary part is zero).
 - `conj(z)` returns `std::complex` even if z is an arithmetic type.
 - `std::linalg` uses *conj-if-needed*(z), which preserves the type of its input.
- For a rank-2 mdspan A :
 - the *transpose* of A is a rank-2 mdspan B such that $B[c, r]$ equals $A[r, c]$; and
 - the *conjugate transpose* of A is a rank-2 mdspan B such that $B[c, r]$ equals *conj-if-needed*($A[r, c]$).
- *BLAS* (pronounced “blahz”) stands for the “Basic Linear Algebra Subroutines,” a Standard Fortran and C interface providing linear algebra operations. This is the foundation of `std::linalg`.

§ 4.2 Common practice: “conjugate transpose of a noncomplex matrix is just the transpose”

The conjugate transpose of a complex matrix naturally generalizes the transpose of a noncomplex matrix. Users who develop generic algorithms for either complex or noncomplex problems write the algorithm once using the conjugate transpose. BLAS and matrix-oriented programming languages like Matlab treat both using the same notation (e.g., the 'C' flag means transpose for a noncomplex matrix, and conjugate transpose for a complex matrix).

§ 4.3 conjugated, transposed, and conjugate_transposed views

A key feature of linear algebra libraries is their ability to view the transpose or conjugate transpose of a matrix “in place” without actually changing its elements. Matrices may be large and copying them may be too expensive.

- BLAS implements “view (conjugate) transpose in place” with a separate flag argument: 'N', 'T', or 'C'.
- `std::linalg` implements this using the `mdspan` view creation functions `conjugated`, `transposed`, and `conjugate_transposed`.

§ 4.4 How does `conjugated` work currently?

1. If the input `mdspan` has accessor type `conjugated_accessor<NestedAccessor>`, then the result has accessor type `NestedAccessor`;
2. otherwise, if the input `mdspan` has accessor type `Accessor`, then the result has accessor type `conjugated_accessor<Accessor>`.

`conjugated_accessor`’s (read-only) access function conjugates the element if it’s a complex number, else it just returns the number.

§ 4.5 This is correct, but can hinder optimization

The current behavior of `conjugated` is mathematically correct, but may result in poor performance.

The problem is that `conjugated(A)` for an `mdspan`-of-noncomplex-numbers `A` should just return `A`, but instead it returns an `mdspan` with a different accessor type than `A`.

This is bad because both Standard Library implementations and users may want to optimize for “known accessors” such as `default_accessor`. Accessors communicate optimization information, like “this is a contiguous array in memory.” Optimizations for known accessors include calling really fast libraries that exploit low-level hardware features. The generic accessor code path may be asymptotically slower in terms of the number of memory accesses.

```
template<class ElementType, class IndexType, size_t Ext0, class
Layout, class Accessor>
void generic_algorithm( // fully generic
    mdspan<ElementType, extent<IndexType, Ext0>, Layout, Accessor>
x);

template<class ElementType, class IndexType, size_t Ext0, class
Layout>
void generic_algorithm( // specialization
    mdspan<ElementType, extent<IndexType, Ext0>, Layout,
default_accessor<ElementType>> x);
```

Currently, `conjugated` of a `default_accessor<ElementType>` `mdspan` has accessor `conjugated_accessor<default_accessor<ElementType>>`. Calling `generic_algorithm` with this `mdspan` will thus take the “generic path,” rather than the specialization.

If we want to optimize the `conjugated_accessor` case, we have to add another specialization. This has compile-time costs. Users either have to remember to do this, or write their generic algorithms twice (once for complex and once for noncomplex).

```
template<class Real, class IndexType, size_t Ext0, class Layout>
requires(not impl::is_complex_v<Real>)
void generic_algorithm( // another specialization
    mdspan<Real, extent<IndexType, Ext0>, Layout,
    conjugated_accessor<default_accessor<Real>>> x)
{
    // Dispatch to default_accessor specialization
    return generic_algorithm(mdspan{x.data_handle(), x.mapping(),
    x.accessor().nested_accessor()});
}
```

§ 4.6 Fix: `conjugated(A)` should return `A` if `A` is noncomplex

P3050 proposes the only reasonable fix: make `conjugated(A)` return `A` if the elements of `A` are not complex.

§ 5 Design justification

§ 5.1 Introduction

LWG finished its review of P1673 at the Kona 2023 WG21 meeting. One reviewer (see Acknowledgments) pointed out that `linalg::conjugated` could be optimized by having it be the identity function if *conj-if-needed* would have been the identity function anyway on the input `mdspan`'s `value_type`. This paper proposes that change. Specifically, if an `mdspan` object `x` has noncomplex `value_type`, and if that `mdspan` does not already have accessor type `conjugated_accessor<A>` for some nested accessor type `A`, then we propose to change `conjugated(x)` just to return `x`.

This change has two observable effects.

1. The result's accessor type will be different. Instead of being `conjugated_accessor<A>` for some `A`, it will just be `A`.
2. If `x` has noncomplex `value_type`, then `conjugated(x)` will no longer have `const element_type`.

We consider Effect (2) acceptable for two reasons.

- a. *in-vector*, *in-matrix*, and *in-object* already do not need to have `const element_type`. Users can pass in views-of-nonconst `mdspan` as read-only vector or matrix parameters. Thus, making the `element_type` of `conjugated(x)` nonconst would not break existing calls to `linalg` functions that take input vector or matrix parameters.

- b. `conjugated(conjugated(z))` for `z` with nonconst complex `element_type` already has nonconst `element_type`. Thus, generic code that depends on the `element_type` of the result of `conjugated` already cannot assume that it is `const`.

§ 5.2 Current behavior of `conjugated`

Currently, `conjugated` has two cases.

1. If the input has accessor type `conjugated_accessor<NestedAccessor>`, then the result has accessor type `NestedAccessor`;
2. otherwise, if the input has accessor type `A`, then the result has accessor type `conjugated_accessor<A>`.

This is correct behavior for any valid `value_type`, because `conjugated_accessor::access` uses *conj-if-needed* to conjugate each element. The exposition-only helper function object *conj-if-needed* uses namespace-unqualified `conj` if it can find it via argument-dependent lookup; otherwise, it is just the identity function. As P1673 explains, *conj-if-needed* exists for two reasons.

1. It preserves the type of its input (unlike `std::conj`, which returns `complex<T>` if the input is a floating-point type and therefore noncomplex).
2. It lets the library recognize user-defined types as complex numbers, as long as `conj` can be found for them via argument-dependent lookup.

The as-if rule would let `conjugated_accessor::access` skip calling *conj-if-needed* and just dispatch to its nested accessor if *conj-if-needed* would have been the identity anyway. However, the accessor type of the `mdspan` returned from `conjugated` is observable, so implementations cannot avoid using `conjugated_accessor`.

§ 5.3 Why change the current behavior?

The current behavior of `conjugated` is correct. The issue is that `conjugated` throws away the knowledge that its input `mdspan` views noncomplex elements. P1673 functions can optimize internally by using `conjugated_accessor::nested_accessor` to create a new `mdspan` for noncomplex `element_type`. However, that costs build time, increases the testing burden, and adds tedious boilerplate to every P1673 function.

This issue also increases the complexity of users' code. For example, users may reasonably assume that if they are working with noncomplex numbers and matrices that live in memory, then they only need to specialize their functions to use `default_accessor<ElementType>`. Such users will find out via build errors that `conjugated(x)` uses `conjugated_accessor` instead. Users may have to pay increased build times and possible loss of code optimizations for this complexity, especially if they write their own computations that use the result of `conjugated` directly as an `mdspan`.

As discussed in P1673 (see the section titled “Why users want to ‘conjugate’ matrices of real numbers”), linear algebra users commonly write algorithms that work for either real or complex numbers. The BLAS assumes this: e.g., DGEMM (Double-precision General Matrix-matrix Multiply) treats `TRANSA='C'` or `TRANSB='C'` ('Conjugate Transpose' in full) as indicating the transpose (same as `'T'` or `'Transpose'`). The Matlab software package uses a trailing single quote, the normal syntax for transpose in Matlab’s language, to indicate the conjugate transpose if its argument is complex, and the transpose if its argument is real. Thus, we expect users to write algorithms that use `conjugate_transposed(x)` or `conjugated(transposed(x))`, even if those users never use complex number types or custom accessors. The current behavior means that such users will need to make their functions’ overload sets generic on accessor type. This proposal would let those users ignore `conjugated_accessor` if they never use complex numbers.

§ 5.4 P1673 layouts and accessors are not “just tags”

Even though we propose to change the behavior of `conjugated`, `conjugate_accessor` needs to retain its current behavior. A key design principle of P1673 is that

... each `mdspan` parameter of a function behaves as itself and is not otherwise “modified” by other parameters.

P1673’s nonwording section “BLAS applies UPL0 to original matrix; we apply Triangle to transformed matrix” gives an example of the application of this principle.

Another way to say that is that the layouts and accessors added by P1673 are not “tags.” That is, P1673’s algorithms like `matrix_product` ascribe no special meaning to `layout_transpose`, `conjugated_accessor`, or `scaled_accessor`, other than their normal meaning as a valid `mdspan` layout or accessors. P1673 authors definitely intended for implementations to optimize for the new layouts and accessors in P1673, but a correct implementation of P1673 can just treat the `mdspan` types generically.

§ 5.5 Change: `conjugated(x)` may no longer have `const element_type`

Both `conjugated_accessor` and `scaled_accessor` have `const element_type`, to make clear that they are read-only views. This also avoids confusion about what it means to write to the complex conjugate of an element, or to the scaled value of an element. This proposal would change `conjugated(x)` to return `x` for `x` with `noncomplex value_type` and with accessors other than `conjugated_accessor<A>` for some `A`. As a result, the result of `conjugated(x)` would no longer have `const element_type` if `x` did not have `const element_type`.

We consider this change acceptable for two reasons.

1. *in-vector*, *in-matrix*, and *in-object* already do not need to have `const element_type`. Users can pass in views-of-nonconst `mdspan` as read-only vector or matrix parameters. Thus, making the `element_type` of `conjugated(x)` nonconst would not break existing calls to `linalg` functions that take input vector or matrix parameters.
2. `conjugated(conjugated(z))` for `z` with nonconst complex `element_type` already has nonconst `element_type`. Thus, generic code that depends on the `element_type` of the result of `conjugated` already cannot assume that it is `const`.

Regarding Reason (2), the current behavior of `conjugated` for an input `mdspan` object `x` with nonconst complex `element_type` is that

- `conjugated(x)` has `const element_type`, but
- `conjugated(conjugated(x))` has nonconst `element_type`.

This proposal would not change that behavior. The following example illustrates.

```
constexpr size_t num_rows = 10;
constexpr size_t num_cols = 11;
vector<complex<float>> x_storage(num_rows * num_cols);

// mdspan with nonconst complex element_type
mdspan<complex<float>,
    dextents<size_t, 2>, layout_right,
    default_accessor<complex<float>>> x{
    x_storage.data(), num_rows, num_cols
};

// conjugated(x) has const element_type,
// because `conjugated_accessor` does.
auto x_conj = conjugated(x);
static_assert(is_same_v<
    decltype(x_conj),
    mdspan<
        const complex<float>, // element_type
        dextents<size_t, 2>, layout_right,
        conjugated_accessor<default_accessor<complex<float>>>
        >
    >);
// x_conj retains the original nested accessor and data handle,
// even though these are both nonconst.
static_assert(is_same_v<
    remove_cvref_t<decltype(x_conj.accessor().nested_accessor())>,
    default_accessor<complex<float>>
>);
// The data handle being nonconst means that we'll be able to
// create conjugated(x_conj), even though conjugated(x_conj)
// has nonconst data handle.
static_assert(is_same_v<
    decltype(x_conj.data_handle()),
    complex<float>*
```



```

>);
// You can't modify the elements through x_conj, though,
// because the reference type is complex<float>,
// not complex<float>&.
static_assert(is_same_v<
    decltype(x_conj)::reference,
    complex<float>
>);

// x_conj_conj = conjugated(conjugated(x));
auto x_conj_conj = conjugated(x_conj);
// x_conj_conj has x's original nested accessor type.
static_assert(is_same_v<
    remove_cvref_t<decltype(x_conj_conj.accessor())>,
    default_accessor<complex<float>>
>);
// That means its element_type is nonconst, ...
static_assert(is_same_v<
    decltype(x_conj_conj)::element_type,
    complex<float>
>);
// ... its data_handle_type is pointer-to-nonconst, ...
static_assert(is_same_v<
    decltype(x_conj_conj.data_handle()),
    complex<float>*>
>);
// ... and its reference type is nonconst as well.
static_assert(is_same_v<
    decltype(x_conj_conj)::accessor_type::reference,
    complex<float>&
>);

```

§ 5.6 What if the input mdspan has conjugated_accessor with noncomplex element_type?

What should `conjugated(x)` do if `x` has accessor type `conjugated_accessor`, but noncomplex `element_type`? The current behavior already covers this case: just strip off `conjugated_accessor` and restore its nested accessor. This proposal does not change that.

Before this proposal, `conjugated` could produce an `mdspan` with accessor type `conjugated_accessor` but noncomplex `element_type`. The only thing that this proposal changes is that it eliminates any way for `conjugated` to reach this case on its own. Users could only get an `mdspan` like that by constructing an `mdspan` explicitly with `conjugated_accessor`, like this.

```

std::vector<float> x_storage(M * N);
std::mdspan x{x_storage.data(),
    std::layout_right::mapping{M, N},
    std::linalg::conjugated_accessor{std::default_accessor{}}};

```

There's no reason for users to want to do this, but the resulting `mdspan` still behaves correctly. We don't prohibit users from doing this.

§ 6 Implementation

This proposal is implemented as [PR 268](#) in the reference `mdspan` implementation.

§ 7 Acknowledgments

Thanks to Tim Song (t.canens.cpp@gmail.com, Jump Trading) for making this suggestion during LWG review of P1673. We have his permission to acknowledge him by name for an LWG review contribution.

§ 8 Wording

Text in blockquotes is not proposed wording, but rather instructions for generating proposed wording.

In [version.syn], for the following definition,

```
#define __cpp_lib_linalg YYYYMMML // also in <linalg>
```

adjust the placeholder vlaue YYYYMMML as needed so as to denote this proposal's date of adoption.

Change [linalg.conj.conjugated] paragraphs 1 and 2 to read as follows. (Paragraph 1 has been reorganized from a sentence into four bullet points, where the old Paragraph 2.2 has been changed to 2.4, and the new Paragraphs 1.2 and 1.3 have been inserted as the middle bullet points. Similarly, Paragraph 2 has been reorganized from two bullet points into four. The old Paragraph 2.2 has been changed to 2.4, and the new Paragraphs 2.2 and 2.3 have been inserted as the middle bullet points.)

1 Let A be

- (1.1) — `remove_cvref_t<decltype(a.accessor().nested_accessor())>` if `Accessor` is a specialization of `conjugated_accessor`; otherwise,
- (1.2) — `Accessor` if `remove_cvref_t<ElementType>` is an arithmetic type; otherwise,
- (1.3) — `Accessor` if the expression `conj(E)` is not valid for any subexpression `E` whose type `T` is expression-equivalent to `remove_cvref_t<ElementType>` with overload resolution performed in a context that includes the declaration `template<class T> conj(const T&) = delete;`; otherwise,
- (1.4) — `conjugated_accessor<Accessor>`.

2 *Returns:*

- (2.1) — `mdspan<typename A::element_type, Extents, Layout, A>(a.data_handle(), a.mapping(), a.accessor().nested_accessor())` if `Accessor` is a specialization of `conjugated_accessor`; otherwise,
- (2.2) — `a` if `remove_cvref_t<ElementType>` is an arithmetic type; otherwise,
- (2.3) — `a` if the expression `conj(E)` is not valid for any subexpression `E` whose type `T` is expression-equivalent to `remove_cvref_t<ElementType>` with overload resolution performed in a context that includes the declaration `template<class T> conj(const T&) = delete;`; otherwise,
- (2.4) — `mdspan<typename A::element_type, Extents, Layout, A>(a.data_handle(), a.mapping(), conjugated_accessor(a.accessor()))`.